

**SMART INDIA
HACKATHON
2025**

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AgroAura is a next-generation AI-powered platform designed to support small-scale farmers and urban gardeners through personalized, accessible, and real-time agricultural support. With features such as a visual BioTwin, multilingual chatbot, disease diagnosis, government scheme access, and market assistance, AgroAura is developed to be a complete, user-friendly digital assistant that works using voice, photo, and manual inputs.

KEY FEATURES



Crop Planner: Provides crop suggestions with detailed cost breakdown, ROI estimation, and government subsidy info.



BioTwin: Virtual crop twin with animated growth, daily care reminders, and step-by-step guidance for farmers.



Disease Detection: AI identifies plant diseases from photos and suggests both organic and chemical treatment options.



Activity Tracking: Logs sowing dates, irrigation schedules, and farm activities for better planning and record-keeping.



Offline Mode: Activities and photos are stored in a local database and sync back automatically once internet is available.

Smart Agents

-  **Planner Agent** – Suggests best crops & plans resources.
-  **Risk Agent** – Warns about pest outbreaks & weather threats.
-  **Market Agent** – Predicts price trends & suggests buyers.
-  **Reminder Agent** – Sends timely care alerts.
-  **Info Agent** – Handles any farming-related queries.
-  **Scheme Agent** – Identifies applicable government benefits.

INNOVATION



Voice-first
AI assistant



Visual BioTwin
for easy learning

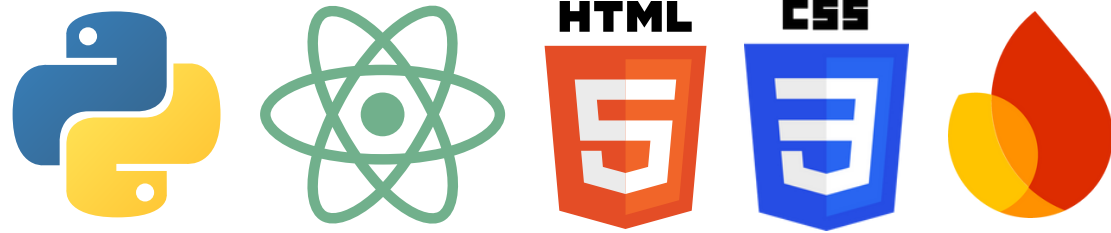


Disease detection
& gov. schemes



Smart Agents
for planning, risk,
market

Programming Languages



Tech Stack

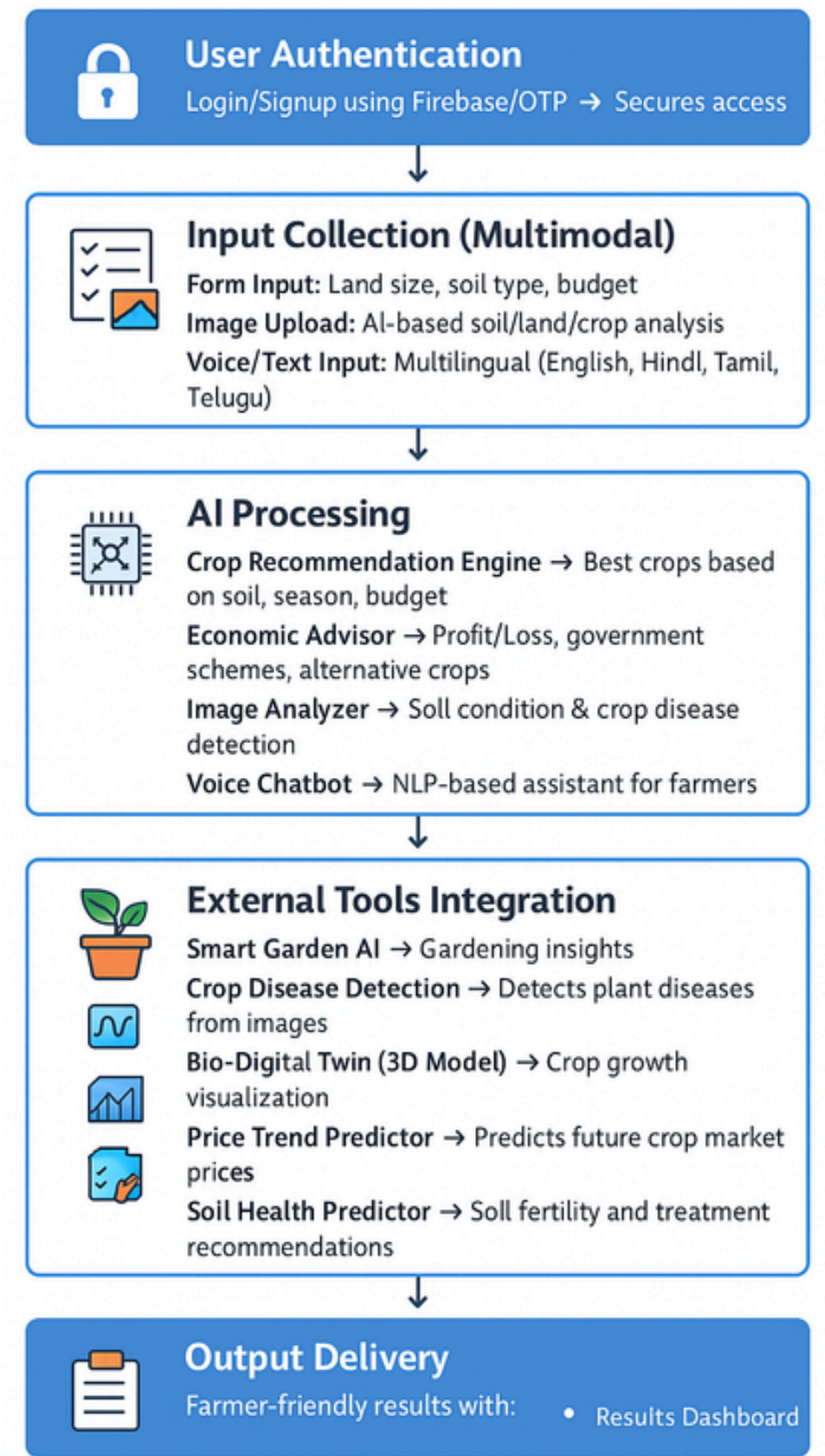
- **Python** → AI/ML models, backend logic, data processing
- **React.js** → Interactive 3D visualization (Digital Twin)
- **HTML5, CSS3** → Web UI
- **Streamlit** → Farmer-friendly web dashboard
- **PyTorch** → AI model (crop recommendation, disease detection & price prediction)
- **OpenCV + PIL** → Image analysis for crop/soil health
- **Transformers (HuggingFace)** → NLP for voice/text queries
- **Whisper ASR** → Speech-to-text voice input
- **Matplotlib** → Graphs & visualizations
- **Firebase** → Storing user info, farm inputs & session data

User Interface & Features



Features / Tools	AgroAura (Our Solution)	Existing Agri Apps / Tools
Form-based Crop Recommendation	✓	✗
Image-based Land & Soil Analysis	✓	✗
Voice/Text Input (Multilingual: EN, HI, TA, TE)	✓	✗
AI-driven Crop Suggestion (soil, land, season, budget)	✓	⚠ Limited
Water & Fertilizer Scheduling	✓	✗
Profit/Loss & Market Economics Planner	✓	✗
Market Price Trend Prediction	✓	✗
3D Bio-Digital Twin (Plant Growth Simulation)	✓	✗
Crop Disease Detection (Image-based)	✓	⚠ Some apps only
Voice-Integrated Chatbot Assistant	✓	✗
Soil Health Prediction (AI-powered)	✓	✗
Government Schemes & Subsidy Mapping	✓	⚠ Manual lookup

Work Flow



Feasibility & Viability

 Technical Feasibility Leveraging existing technologies like machine learning and cloud computing	 Economic Viability Cost-effective solution with potential for high ROI
 Operational Feasibility Easy integration with existing agricultural practices	 Environmental Sustainability Promotes sustainable farming practices and resource optimization

Potential Risks & Mitigation Strategies

 DATA QUALITY & AVAILABILITY Inconsistent, incomplete or outdated data	 SOLUTION Use verified datasets and apply data preprocessing
 TECHNOLOGY ACCESSIBILITY Limited internet or smartphone usage	 SOLUTION Provide offline-first and voice-based support
 DEPENDENCE ON EXTERNAL APIs API downtime or inaccurate data	 DPENDENCE ON EXTERNAL APIs Maintain fallback with cached data

Vulnerabilities & solution

VULNERABILITIES	SOLUTIONS
 TECHNOLOGICAL Hardware/software failures	 Implement redundancy and backup systems
 SECURITY Cyberattacks and data breaches	 Employ robust encryption and security protocols
 API IALTIY API downtime or inaccurate data	Maintain fallback with cached data

BENEFITS**Social Benefits**

- *Empowers farmers*
- *Inclusive & accessible*
- *Supports urban gardening*

Economic Benefits

- *Increases yield & income*
- *Enhances market access*
- *Boosts agricultural GDP*

Environmental Benefits

- *Reduces crop loss*
- *Encourages sustainable practices*
- *Optimizes resource usage*

IMPACT***For Farmers***

- *Better crop yield & productivity*
- *Higher income & profit*
- *Reduced risk of crop failure*

For Consumers

- *Access to healthier & safer food*
- *Stable prices through better supply*
- *Wider variety of fresh produce*

Impacts for Government

- *Better agricultural data for policy-making*
- *Promotes sustainable & smart farming*
- *Enhances farmer welfare & rural development*

RESEARCH AND REFERENCES

- FAO – Digital Agriculture Report (2022) : <https://www.fao.org/digital-agriculture>
 - Springer – Co-Creation in Digital Platforms(2021) : <https://doi.org/10.1007/s10660-022-09574-4>
 - Press Information Bureau – AI in Agriculture : <http://bit.ly/PIB-AI-Agriculture>
 - Harvard – Why Customer Feedback Is Essential(2019) : <http://bit.ly/HBR-Customer-Feedback>
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- Youtube link of Prototype : <https://youtu.be/aLHC5x7D1iA>